Sizing Matters

Position sizing for fundamental PMs [V.0.9]

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Why is this position twice as large?

- A larger expected payoff can justify more dollars.
- Greater uncertainty can justify fewer dollars, even with the same payoff.
- The horizon and the path to the catalyst affect the risk we can carry.
- The rest of the book determines how much room the idea has.

The objective: a dollar target that the PM and analyst can explain, challenge, and update.

Selection and sizing: the benchmark

For each day, let x_i be a signed dollar position and N the number of positions.

$$g = \frac{1}{N}\sum_{j=1}^{N}|x_j|, \qquad x_i^{\text{selection}} = \text{sign}(x_i)g, \qquad x_i^{\text{sizing}} = x_i - x_i^{\text{selection}}.$$

- **Selection:** retain each long/short direction and give every position equal absolute dollars.
- **Sizing:** measure deviations from that equal-dollar portfolio.
- The positions add exactly. Gross exposure is preserved in the selection benchmark, but net and risk exposures can change.

In the supplied summaries, selection and sizing PnLs do not generally add to reported total PnL. Keep the denominators distinct.

Sample: 155 ACUs, each treated as one PM observation, active at some point since 2021, with more than 150 observations and an assigned theme.

Selection and sizing outcomes

Percentages within each group, classified by **reported total PnL**.

Total PnL	% with Sel. > Siz.	% with sizing PnL > 0	% with selection PnL > 0	Median sizing / total PnL (%)	PMs
Nonpositive	45	35	41	—	75
Positive	68	72	89	25	80

Among profitable PMs, sizing PnL is positive for **72%**. The median sizing / reported-total-PnL ratio is **25%**.

Source: anonymized BAM LTD summaries. The nonpositive group includes six zero-total-PnL observations. The median ratio is shown only for profitable PMs.

Sizing contributes 23.1% of combined PnL

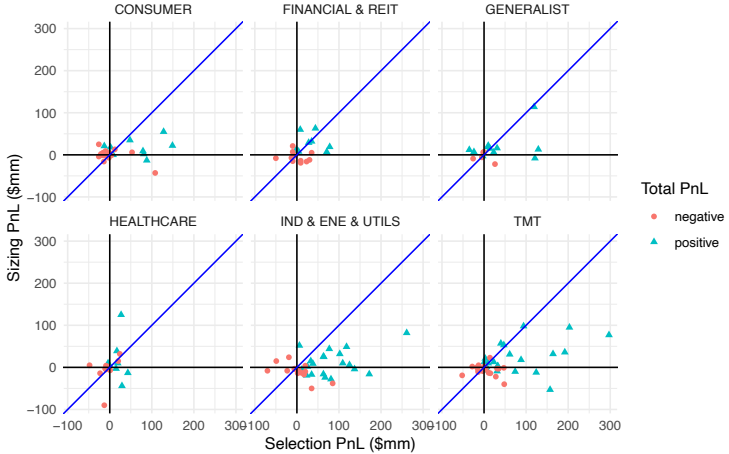
80 profitable PM observations. Sum each component, then calculate its share.

Theme	Selection share (%)	Sizing share (%)	PMs
CONSUMER	79.3	20.7	9
FINANCIAL & REIT	52.6	47.4	9
GENERALIST	65.0	35.0	11
HEALTHCARE	52.8	47.2	9
IND & ENE & UTILS	88.4	11.6	21
TMT	78.2	21.8	21
ALL	**76.9**	**23.1**	**80**

Selection contributes about **3.3 times** as much as sizing in aggregate. Sizing's contribution remains material.

Source: anonymized BAM LTD summaries. Denominator: combined selection + sizing PnL, not reported total PnL. These are dollar-weighted aggregate shares, not the median PM's shares.

Selection and sizing PnL by theme



One point per PM. Blue line: equal PnL. "Negative" includes zero total PnL.

Correlations across PM outcomes

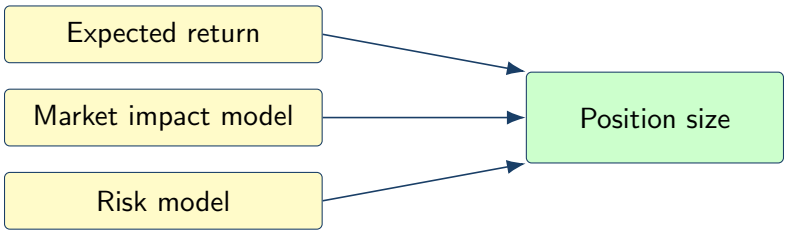
Cross-sectional correlations of **LTD dollar PnLs**, all 155 PM observations.

	Total PnL ($mm)	**Alpha PnL ($mm)**	**Selection PnL ($mm)**	**Sizing PnL ($mm)**
Total PnL ($mm)	1.00	0.91	0.80	0.61
Alpha PnL ($mm)	0.91	1.00	0.90	0.67
Selection PnL ($mm)	0.80	0.90	1.00	0.34
Sizing PnL ($mm)	0.61	0.67	0.34	1.00

Strong selection outcomes do not automatically come with strong sizing outcomes. Their cross-sectional correlation is **0.34**.

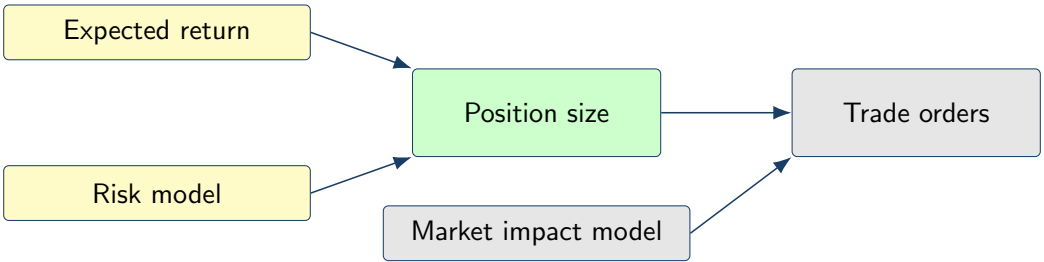
Source: anonymized BAM LTD summaries. These correlations measure neither profit shares nor diversification through time.

A first picture of position sizing



- This picture combines the target position and the mechanics of trading.
- It also leaves forecast uncertainty and the investment horizon implicit.

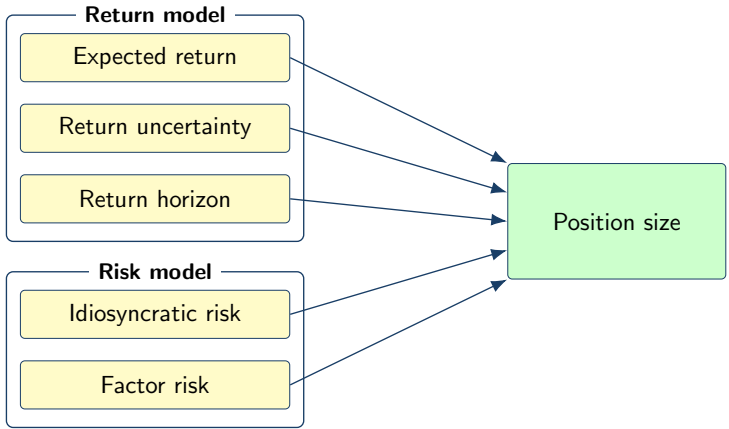
Target positions and trade orders



The PM sets the intended exposure. Execution determines the timing and size of orders used to reach it.

Liquidity, capacity, expected costs, and time to exit still constrain a feasible target. Execution mechanics do not remove these PM responsibilities.

The inputs to a position target



Return uncertainty means uncertainty about the expected-return estimate. It is distinct from volatility around that unknown expectation.

The fundamental disagreement

An illustrative earnings case

Team EPS forecast	\$1.60
Reference EPS estimate	\$1.10
Team EPS without a planned price increase	\$1.35

The earnings disagreement survives this stress. The price increase still needs investigation.

The bridge to a stock return

- Identify the reporting period and the operating driver.
- Explain what the market already expects.
- Translate earnings into a share-price payoff under a stated valuation.
- Identify the catalyst and timing.

A larger earnings difference is useful only if it produces a credible remaining payoff.

What does conviction mean?

Expected return, α_i

What payoff do we expect, over which horizon?

Forecast standard error, τ_i

How uncertain are we about that expectation?

Evidence behind confidence

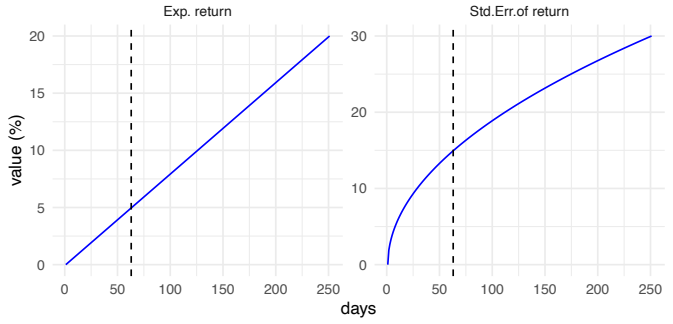
- Contractual or accounting facts versus demand assumptions.
- Sensitivity to the largest drivers.
- Management competence and transparency, assessed separately.

A $+10\%$ forecast with uncertainty of 50% of its magnitude has $\tau_i = 5$ **percentage points**, not 50 percentage points.

A qualitative conviction score does not automatically produce a calibrated standard error. Record assumptions and test sensitivity.

Forecasts on a common horizon

Illustrative convention: expected return scales with time, forecast standard error with its square root.



$$\alpha_{\text{quarter}} = 20\% \times \tfrac{1}{4} = 5\%, \qquad \tau_{\text{quarter}} = 30 \text{ pp} \times \sqrt{\tfrac{1}{4}} = 15 \text{ pp}.$$

Illustration from the sizing note. Dashed line: 63 trading days. Square-root scaling requires an uncertainty model. Event-driven forecasts require explicit catalyst timing.

Low volatility magnifies forecast errors

For uncorrelated company-specific returns, known expected returns:

$$x_i = \kappa \frac{\alpha_i}{\sigma_i^2}, \qquad \kappa > 0.$$

Same expected return

$$\sigma_A = 15\%, \quad \sigma_B = 45\%$$

$$\frac{x_A}{x_B} = \left(\frac{45}{15}\right)^2 = 9.$$

Same forecast error

$$\Delta x_i = \kappa \frac{\Delta \alpha_i}{\sigma_i^2}$$

At fixed scale, an equal error changes the quieter stock's position nine times as much.

Low return volatility does not establish a precise expected-return forecast.

x_i: signed dollar position. α_i: expected idiosyncratic return. σ_i: idiosyncratic return volatility. All inputs use the same holding period.

Sizing with forecast uncertainty

$$x_i = \kappa \frac{\alpha_i}{\sigma_i^2 + \tau_i^2}$$

σ_i^2: return variance

Fluctuations around the unknown expected return.

τ_i^2: forecast variance

Uncertainty about the expected return itself.

- Use a common horizon and consistent return units.
- Assume independent forecast uncertainty and return noise. Count each once.
- The stock-by-stock rule assumes no cross-stock correlations in either source.

Use the rule to establish relative targets. Set overall portfolio risk and impose constraints separately.

Sizing rules and the information they use

Rule	Dollar position x_i	Interpretation
Mean–variance	$\kappa \frac{\alpha_i}{\sigma_i^2}$	Expected return and idiosyncratic variance. Treats expected return as known.
Alpha over volatility	$\kappa \frac{\alpha_i}{\sigma_i}$	Equal standalone dollar volatility for equal absolute forecasts.
Uncertainty-adjusted	$\kappa \frac{\alpha_i}{\sigma_i^2 + \tau_i^2}$	Includes uncertainty about expected returns in the variance penalty.
Uncertainty dominates	$\kappa \frac{\alpha_i}{\tau_i^2}$	Approximation when $\tau_i^2 \gg \sigma_i^2$.
Proportional	$\kappa \alpha_i$	Dominant forecast uncertainty is approximately common across stocks.

α_i: expected idiosyncratic return. σ_i: idiosyncratic volatility. τ_i: forecast standard error. $\kappa > 0$ is common within each rule and calibrated separately for each rule. Same holding period throughout.

How uncertainty changes a dollar position

One-year $\alpha = 10\%$. Each rule starts at \$10m for $\sigma = 30\%$, $\tau = 0$.
Hold each rule's own scale fixed as the assumptions change. Positions below are **\$m**.

Idiosyncratic volatility	Forecast standard error	Mean–variance	Alpha over volatility	Uncertainty-adjusted
15%	0%	40.00	20.00	40.00
30%	0%	10.00	10.00	10.00
45%	0%	4.44	6.67	4.44
15%	30%	40.00	20.00	8.00
30%	30%	10.00	10.00	5.00
45%	30%	4.44	6.67	3.08
15%	90%	40.00	20.00	1.08
30%	90%	10.00	10.00	1.00
45%	90%	4.44	6.67	0.89

Illustrative scenarios from the sizing note. A 90% standard error means 90 percentage points and is deliberately extreme. Portfolio risk rescaling occurs after this comparison.

When proportional sizing is a useful approximation

Two conditions: forecast variance dominates return variance, and absolute forecast uncertainty is approximately common across stocks.

$$\tau_i^2 \gg \sigma_i^2, \qquad \tau_i \approx \tau \quad \Longrightarrow \quad x_i \approx \frac{\kappa}{\tau^2}\alpha_i.$$

- Twice the expected return then receives roughly twice the dollars.
- Equal absolute dollars follow only from equal absolute forecasts.
- Evidence of different forecast precision belongs in the full denominator.

Common uncertainty in percentage points differs from common uncertainty relative to each forecast. A common relative error does not generally justify this approximation.

The loss budget places the target in dollars

Illustrative long position

Dollar loss budget: **$500,000**

Adverse price move: **25%**

$$\text{Position} = \frac{\$500{,}000}{0.25} = \boxed{\$2 \text{ million}}.$$

The path must be tolerable

- Consider the event and the period before resolution.
- Account for other positions that could lose at the same time.
- Check exit liquidity and short-squeeze scenarios where relevant.

The scenario informs the allocation. It does not guarantee that realized losses stop at the budget.

The portfolio determines the final exposure

1. **Compute relative sizes** using company-specific forecasts, risk, and forecast uncertainty on a common horizon.
2. **Choose one portfolio scale** using the full risk model, including correlations, to fit the volatility budget.
3. **Use the RD optimizer** to control factor exposures and apply portfolio constraints.
4. **Recheck the result** for concentration, related companies, scenario losses, and liquidity. Recompute risk after adjustments.

Two names can be one economic bet. Common business drivers and common forecasting mistakes both matter.

Size changes as the investment case changes

Remaining payoff and catalysts

- Recalculate upside and downside after price moves or new evidence.
- Map earnings, the investor day, and later events.
- Compare the next event with the longer-term earnings case.

A long falls 5%

An unchanged earnings case may leave a better entry price.

A lost customer may invalidate the forecast.

The same price move can call for a different sizing decision.

Reassess the thesis and remaining loss budget before considering an addition. Willingness to add does not validate the forecast.

A shared record connects research to holdings

The investment record

- Forecast date and horizon.
- Expected return, upside/downside, and forecast uncertainty with units.
- Operating model, evidence, catalyst, and invalidation conditions.
- Open research questions and model version.

The position decision

- Target dollars and actual dollars.
- Loss budget and portfolio constraints.
- Reasons for an override or staged change.
- Next review and what could change the target.

Research scores and portfolio ranks serve different purposes. Equally ranked ideas can require different dollar exposures.

A sizing checklist for the PM review

1. **Forecast:** state the company-specific payoff and horizon.
2. **Uncertainty:** document the evidence, units, and sensitivity.
3. **Relative size:** apply the uncertainty-adjusted rule with its assumptions.
4. **Portfolio:** scale to the budget, constrain exposures in RD, and recheck risk.
5. **Sanity checks:** assess loss scenarios, concentration, related names, and liquidity.
6. **Learning:** update after material news and review sizing outcomes quarterly.

Explain the target before the outcome. Review the outcome against the explanation that was recorded at the time.

Discussion

What would change the size of your largest idea?

- A revised payoff or a different catalyst horizon?
- Weaker evidence behind a key assumption?
- Less room in the portfolio's loss or risk budget?
- A target that has drifted away from the actual holding?

Appendix: attribution and evidence

- **Population.** 155 filtered ACUs active at some point since 2021, each with more than 150 observations and an assigned theme. Each ACU counts as one PM observation.
- **Profitability.** 80 positive and 75 nonpositive reported-total-PnL observations. The latter include six zeros.
- **Two statistics.** 23.1% is the aggregate sizing share of combined component PnL among profitable PMs. 25% is their median sizing / reported-total-PnL ratio.
- **Benchmark.** Equal absolute GMV retains stock directions and gross dollars. It can change net, factor, and risk exposures.
- **Limits.** Components do not generally reconcile to reported total. Cumulative outcomes mix scale and track-record length. Correlations are across PMs.

The evidence describes realized outcomes. It establishes neither persistent sizing skill nor the causal benefit of a proposed sizing rule.

Appendix: the uncertainty-adjusted rule

Predictive model. $\mathbf{r} = \mathbf{m} + \boldsymbol{\epsilon}$, with

$$E[\mathbf{m}] = \boldsymbol{\alpha}, \quad \mathrm{Cov}(\mathbf{m}) = T, \quad E[\boldsymbol{\epsilon}] = 0, \quad \mathrm{Cov}(\boldsymbol{\epsilon}) = S, \quad \mathbf{m} \perp \boldsymbol{\epsilon}.$$

Then $E[\mathbf{r}] = \boldsymbol{\alpha}$ and $\mathrm{Cov}(\mathbf{r}) = S + T$.

$$\max_{\mathbf{x}} \left\{ \boldsymbol{\alpha}^\top \mathbf{x} - \frac{1}{2\kappa} \mathbf{x}^\top (S + T)\mathbf{x} \right\}, \qquad \kappa > 0.$$

For $S + T \succ 0$, the unconstrained solution is

$$\boxed{\mathbf{x} = \kappa (S + T)^{-1} \boldsymbol{\alpha}}.$$

If S and T are diagonal, $x_i = \kappa \alpha_i / (\sigma_i^2 + \tau_i^2)$.

The covariance must represent distinct uncertainty sources. Constraints change the solution. A diagonal approximation omits common forecast errors.

Appendix: horizons and uncertainty units

The chart's convention, for a horizon fraction h:

$$\alpha_h = h\alpha_1, \qquad \tau_h = \sqrt{h}\,\tau_1.$$

Independent accumulation of forecast errors can motivate this convention. Uncertainty about a persistent drift instead scales with h.

Two ways to report the same standard error:

$$\alpha = \pm 10\%, \quad c = 50\% \text{ of } |\alpha| \quad \Longrightarrow \quad \tau = c|\alpha| = 5 \text{ percentage points.}$$

- Common *absolute* τ, when dominant: $x_i \propto \alpha_i$.
- Common *relative* c, when dominant and $\alpha_i \neq 0$: $x_i \propto 1/\alpha_i$.
- At zero forecast, specify uncertainty in absolute units. Near zero, the dominance approximation can fail.

The inverse-alpha result is a limiting implication. Use the full denominator for sizing, with a stated horizon model and consistent uncertainty units.

Appendix: the research foundation

1. **Record forecasts and open questions.** Use a shared format linked to the financial model, with dates, horizons, and evidence.
2. **Quantify the disagreement.** State the earnings or cash-flow difference, its operating driver, and the share-price consequence.
3. **Form an independent view.** Build from company evidence before relying on other investors' assumptions.
4. **Protect research time.** Automate routine updates and limit noise that does not change the case.
5. **Build cumulative industry knowledge.** Seek situations where that knowledge helps explain complex company developments.
6. **Connect a detailed operating model to outside evidence.** Reconcile periods, revenue recognition, and what each data source measures.

Appendix: the quality of the evidence

7. **Find what others missed and quantify its importance.** Novel information must materially change the forecast.

8. **Prepare conversations around unanswered questions.** Test mechanisms with management, customers, suppliers, and experts.

9. **Assess competence and transparency separately.** Reconcile management's statements with operating evidence and accounts.

10. **Separate dependable drivers from informed guesses.** Repeated versions of the same uncertain evidence may add little confidence.

11. **Stress the most important drivers.** Identify how much of the earnings disagreement survives disappointment.

12. **Identify exposures the research cannot forecast.** Consider feasible hedges and the risks they leave behind.

Appendix: payoff, timing, and loss tolerance

13. **Update remaining upside and downside.** Link operating outcomes to valuation and include adverse moves around catalysts.

14. **Map the catalyst sequence and invalidation evidence.** Separate a delayed payoff from a broken investment case.

15. **Compare the next event with the longer-term thesis.** Adjust size and holding period when the horizons conflict.

16. **Give every idea an explicit dollar loss budget.** Combine it with an adverse scenario and assess the path to resolution.

17. **Use a shared qualitative checklist.** Define strong and weak evidence, and keep research assessments distinct from sizing ranks.

Appendix: targets, updates, and learning

18. **Set dollar targets across the book.** Explain ranks, risk, and competing opportunities, then review actual holdings and overrides.

19. **Test size against an adverse move.** Reassess the thesis and loss budget before deciding whether adding makes sense.

20. **Refresh inputs at least as often as the investment horizon.** Review sooner when material information arrives.

21. **Learn from every earnings release.** Reconcile the outcome with the model and distinguish timing errors from a changed case.

22. **Review sizing with the team quarterly.** Examine selection and sizing PnL and the largest outcomes against recorded decisions, then improve the checklist.